

Instructor: Date:

Learner: Regional Branch:

Program: Center:

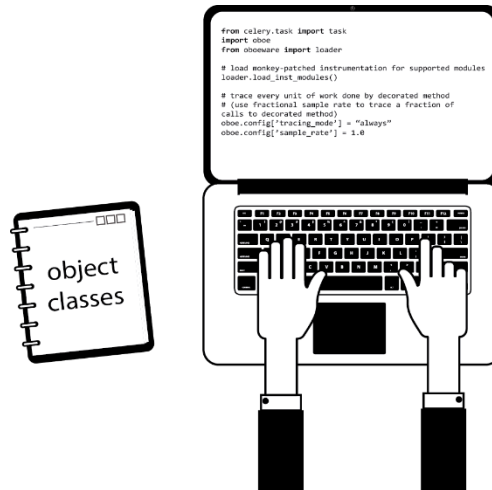
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Let's Start!

Important Aspects in OOP



Learning Outcome:

Talk about principles, concepts, characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- Do you remember what the different uses of object-oriented programming are?
- What is the importance of object in programming languages?

Can Do Statements:

- I can explain advanced concepts related to object-oriented programming.
- I can talk about the importance of classes in object-oriented programming.
- I can give examples about the concepts of method and function in object-oriented programming.





Let's Explore!

Let's listen to this podcast about OOP.



Instructions: Read the text below.

I will talk today about OOP. You might want to know that OOP stands for object-oriented programming, and it is very useful in the programming area as it contributes to problem-solving procedures. You're going to have to know about all aspects related to this topic if you want to become an outstanding programmer, and for that, you ought to learn about 'classes'. Before creating any object, you'd better create its class first as it's basically the outline that the object will have once it's finished. Actually, it is of great help when you're going to create several objects as the same class will be implemented in each object saving you time; this way, you can include the particular characteristics for each one of those later.

Now, you might want to know about the four different concepts that stem from OOP. First, you will have to learn about abstraction, in which the programmer only displays the most relevant information depending on the context. Now, you'd better bear in mind that encapsulation refers to a safety measure that protects data and restricts access to only authorized users. You also ought to know that with inheritance, we can create new classes from old ones, keeping the main characteristics of them. Finally, with polymorphism, we'll have several forms that share the same name.

Other important elements to take into account are going to be 'methods' and 'functions'. A method delimits how an object behaves; for example, a method related to a 'dog' object ought to be barking or running. On the other hand, a function collects input to achieve a final result; for example, a function can calculate the price of a taxi by measuring the distance and time. Anyway, don't forget that I arrive tonight at your city. I'm speaking at the programmers' convention tomorrow at 6 pm!



Let's Practice!

Methods and Functions Features

Instructions: Select the correct answer.

- You're _____ to know about all aspects related to this topic if you want to become an outstanding programmer.
 - having
 - going
 - go
 - going to have
- Other important elements to take into account _____ be 'methods' and 'functions'.
 - is going to
 - are going to
 - is
 - be going to
- A method delimits how an object _____ in specific situations.
 - will behave
 - behaving
 - behave
 - are behaving

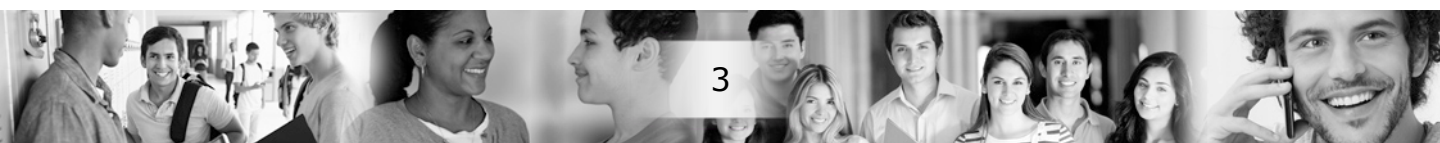


Let's Practice!

Getting to Know Some Tools a Programmer Needs in Order to Be Better at Work

Instructions: Read the text again. Circle the correct word to complete the sentence.

- You might (want to know / need to know / want to make) that OPP stands for object-oriented programming.
- If you want to become an outstanding programmer, you (must learn / shouldn't learn / ought to learn) about 'classes'.
- You'd (better increase / better create / better make) its class first, as it's basically the outline that the object will have once it's finished.
- To create several objects, you (should use / ought to use / has to use) classes as the same class will be implemented in each object.
- You might (want to save / has to save / saving) your time, so you can include the particular characteristics for each class later.





Extension Activity

Explaining the object-oriented and classes with daily examples

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for extra information about object-oriented and classes' examples, read them carefully and take notes.
- Describe each detail of one of them and create a presentation to explain their features.
- Share with your partners and answer their questions.

Self-Assessment

Yes

No

- I look for extra information about object-oriented and classes' examples.
- I create a presentation to explain their features.
- I talk about object-oriented and classes' features and compare with basic examples.

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Target Words

- outstanding** (adjective): successful; one of the best
e.g. You're going to have to know about all aspects related to this topic if you want to become an outstanding programmer.
- outline** (noun): the scheme or map of something
e.g. Before creating any object, you'd better create its class first as it's basically the outline that the object will have once it's finished.
- stem from** (phrasal verb): to originate from
e.g. Now, you may want to know about the four different concepts that stem from OOP.
- relevant** (adjective): extremely important
e.g. First, you will have to learn abstraction, in which the programmer only displays the most relevant information depending on the context.
- safety measure** (noun): a preventive action taken to avoid any accident
e.g. Now, you'd better bear in mind that encapsulation refers to a safety measure that protects data and restricts access to only authorized users.
- delimit** (verb): to define something
e.g. A method delimits how an object behaves.



Learning Points

The Future With Will, Going to, the Present Continuous and the Present Simple

There are many different ways to talk about future plans or events.

WILL

You use **will** to talk about future plans as long as some conditions are met. The use of **will** can be used to talk about an opinion or prediction about the future, a spontaneous decision taken at the moment, or a promise. The use of **will** goes along with expressions like *probably, maybe, perhaps*, among others. Use the following structure:

Subject + will + verb (base form) + complement.

For example:

*She **will probably invest** in our company next year.*

*Gabriel and Mark just made their decision; they **will buy** our software.*

GOING TO

You use **going to** to talk about future plans depending on the situation; when you express a prediction based on the present time or to talk about a plan you have already made the decision to do. Remember to use the verb **to be** right before the expression **going to**. Take a look at the structure we use:

Subject + am/is/are + going to + verb (base form) + complement

For example:

*I **am going to** try that new software today.*

*The meeting **is going to** include the intervention of Mr. Gonzalez, a special engineer of the committee.*

PRESENT CONTINUOUS

You can use the **present continuous** tense to talk about plans in the future; however, the plan you are referring to must be already planned or prepared in some degree. It is necessary to include a time expression that refers to the future. The structure we use is the following:

Subject + am/is/are + action verb in gerund (-ing) + complement

For example:

*We **are meeting** with the new staff of the tech department tomorrow **at 7 o'clock**.*

*The company **is launching** the new product **next summer in July**.*





Learning Points

PRESENT SIMPLE

You can use the **present simple** to talk about situations that are scheduled or already arranged. Apart from using the **present simple** tense, the only requirement in here is to explicitly add the future time phrase that mentions when the situation or event is going to take place.

For example:

*The investors arrive **tonight at 8 pm.***

*The meeting finishes **at 6.30 in the afternoon.***

Examples

- *I will talk today about OOP.*
- *You're going to have to know about all aspects related to this topic if you want to become an outstanding programmer.*
- *Anyway, don't forget that I arrive tonight at your city; I'm speaking at the programmers' convention tomorrow at 6!*

Breaking Down Concepts in Object Oriented Programming

In the engineering field, there are **concepts in OOP** that you have to completely understand. Take a look at the following chart to learn more about the four basic concepts that belong to the **OOP** area.

CONCEPT	DESCRIPTION
Abstraction	This concept refers to the situation in which the programmer only displays the most important information depending on the context.
Encapsulation	This concepts refers to a safety measure programmers can use. It protects data and allows access to authorized users; furthermore, it hides internal mechanisms as well as data structures.
Inheritance	Bear in mind that this concept is a process in which new classes emerge from pre-existing ones, keeping the general characteristics of these. It's a way to reuse codes.
Polymorphism	This concepts refers to the situation in which multiple forms share the same name. Plus, the characteristics that an object has can be modified according to the context.